Halokazi

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Introduction

This document describes the software development methodology adopted for the Halokazi project: a Hybrid SDLC Model that combines the Waterfall approach with Agile Scrum.

Waterfall is used for the initial planning, requirements, analysis and design work, where a clear and well-documented foundation is needed before implementation starts. Agile Scrum is then used for the actual coding and testing of the system, which is broken into short, repeating sprints so that working software can be delivered, reviewed and adjusted incrementally rather than all at once at the end of the project. Once implementation is complete, the project returns to a Waterfall-style approach for system-wide testing, deployment, final documentation and ongoing maintenance.

This approach gives the project the documentation rigor expected of a formally planned system, while keeping development flexible enough to respond to feedback and changing priorities during implementation.

Methodology Overview

WATERFALL Stages 1–9 Planning & Design	AGILE / SCRUM Stage 10 Sprints (repeats)	WATERFALL Stages 11–14 Completion & Maintenance

The project moves through 14 stages in total: nine planning stages under Waterfall, one implementation stage under Agile Scrum that repeats for as many sprints as the backlog requires, and four completion/maintenance stages back under Waterfall. Stage 9 is the formal sign-off point where responsibility passes from Waterfall planning to Agile implementation.

Stage-by-Stage Description

WATERFALL — PLANNING

Stage 1: Project Initiation & Idea Definition

The project idea is formally defined: the problem being solved, the proposed solution, who will use the system, and why it matters. The project's vision, mission, objectives, high-level scope and core modules are documented, giving the whole project a clear direction and boundaries before any detailed work begins.

Key outputs:

- Project Initiation & Idea Definition document
- Vision, mission and objectives
- High-level scope, stakeholders and core modules

Stage 2: Feasibility Study

Before committing time and resources, the project is evaluated to confirm it is realistic to build. This covers technical feasibility (can it be built with available skills and technology), Financial feasibility (cost versus available budget), Operational feasibility (will it be usable and sustainable), Schedule feasibility (can it be completed in the available time), and Legal/Privacy feasibility (data protection and compliance obligations). The stage ends with a clear Go / No-Go recommendation.

Key outputs:

- Feasibility Study Report
- Go / No-Go recommendation

Stage 3: Requirement Gathering & Analysis

Detailed system requirements are collected and documented. Functional requirements describe what the system must do (e.g. register, create CV, search jobs, apply, receive alerts), while non-functional requirements describe how well it must do it (security, performance, usability, scalability). User roles such as Job Seeker, Employer, Administrator are confirmed and validated.

Key outputs:

- Software Requirements Specification (SRS)
- Functional and non-functional requirements list
- Confirmed user roles

Stage 4: System Analysis / Business Analysis

The team builds a clear understanding of how the system should operate before any diagrams or design work starts. This includes mapping the current situation (AS-IS) against the intended future situation (TO-BE), documenting business rules, and identifying actors and high-level use cases. Detailed User Stories are intentionally not written here, only enough detail is captured to inform Modeling and Architecture. Full User Stories and Acceptance Criteria are produced later, in Stage 9, to avoid duplicating the same work twice.

Key outputs:

- AS-IS / TO-BE process documentation
- Business rules
- Actor list and high-level use cases

Stage 5: System Modeling

Requirements and analysis are translated into visual technical models that will guide design and development. This includes Use Case Diagrams, Activity Diagrams, Sequence Diagrams, an Entity-Relationship Diagram (ERD), a Context Diagram, Data Flow Diagrams (DFD), Class Diagrams, and a System Architecture Diagram.

Key outputs:

- Complete set of UML / system diagrams

Stage 6: Database Design

Based on the ERD and requirements, the database structure is designed in detail as tables, fields, data types, primary/foreign keys, relationships and normalization. Core tables typically include users, profiles, cvs, cv_templates, jobs, companies, applications, job_alerts and notifications.

Key outputs:

- Database Design Document
- Finalized schema and data dictionary

Stage 7: UI/UX Design

Wireframes and mockups are created for all major screens including dashboards, CV builder, job listings, application flow, employer and admin portals… defining layout, navigation and user experience before any coding starts.

Key outputs:

- Wireframes and mockups for all key screens

Stage 8: Technology and Architecture Selection

The technology stack is decided: frontend framework, backend framework, database engine, authentication method, notification service, file storage and deployment platform. The overall system architecture (monolithic vs. API-based) is also confirmed here.

Key outputs:

- Technology stack decision document
- Architecture diagram / decision record

SIGN-OFF: TRANSITION TO AGILE

Stage 9: Development Planning → Product Backlog

This is the bridge between Waterfall and Agile. The high-level requirements and analysis produced in Stages 3 and 4 are now formally broken down into Epics, then Features, then detailed User Stories with Acceptance Criteria, and prioritized into a Product Backlog ready for sprint planning. Everything completed up to this point (Stages 1–9) must be reviewed and signed off before implementation begins - this is the official handover point from Waterfall planning to Agile execution.

Key outputs:

- Product Backlog (Epics → Features → User Stories → Acceptance Criteria)
- Sign-off approval to begin implementation

AGILE / SCRUM: IMPLEMENTATION

Stage 10: Sprint Cycles (Agile / Scrum Development)

Actual coding takes place here, in repeating iterations called sprints (typically 1–4 weeks each). Each sprint follows the same short cycle:

Sprint Planning (select backlog items for this sprint) → Design (detailed design for that increment) → Development (coding) → Testing (Unit, API and Feature testing performed inside the sprint) → Sprint Review/Demo (show completed, working functionality to stakeholders) → Retrospective (the team reflects on what went well and what to improve).

The cycle then repeats with the next sprint until the backlog is complete. Modules such as Authentication, CV Builder, Job Search, Applications, Notifications, Employer Portal and Admin Dashboard are typically each built across one or more sprints.

Key outputs:

- Working, tested software increment after every sprint
- Updated / re-prioritized backlog

WATERFALL: COMPLETION

Stage 11: System-Wide Testing

Once the major sprints are complete, the system is tested as a whole rather than module-by-module. This includes Full Integration Testing (modules working correctly together), System Testing, Security Testing, Performance Testing, User Acceptance Testing (UAT) with real stakeholders, and Regression Testing to confirm that earlier sprints were not broken by later changes.

Key outputs:

- Test Plan and Test Cases
- Test Report / defect log
- UAT sign-off

Stage 12: Deployment

The system is moved into a live production environment. This covers domain setup, SSL certificate installation, production database setup, environment variables and configuration, activation of the notification service, backup configuration, and logging and monitoring setup.

Key outputs:

- Live production system
- Deployment and Configuration Documentation

Stage 13: Final Documentation

All documentation produced throughout the project is consolidated, updated to reflect any changes made during the sprints, and finalized — including the SRS, System Design Document, Database Documentation, API Documentation, User Manual and Administrator Manual.

Key outputs:

- Final SRS and System Design Document
- API Documentation
- User Manual and Admin Manual

ONGOING: POST-LAUNCH

Stage 14: Maintenance & Continuous Improvement

After launch, the project moves into ongoing support: fixing bugs, applying security updates, monitoring performance and uptime, running regular backups, and adding new features based on real user feedback. This stage does not have a fixed end date — it continues for as long as the platform is in operation.

Key outputs:

- Bug-fix and update log
- Monitoring / uptime reports
- Continuous backlog of future improvements

Quick-Reference Summary Table

#	Stage	Phase	Key Output
1	Project Initiation & Idea Definition	Waterfall - Planning	Project Initiation & Idea Definition document
2	Feasibility Study	Waterfall - Planning	Feasibility Study Report
3	Requirement Gathering & Analysis	Waterfall - Planning	Software Requirements Specification (SRS)
4	System Analysis / Business Analysis	Waterfall - Planning	AS-IS / TO-BE process documentation
5	System Modeling	Waterfall - Planning	Complete set of UML / system diagrams
6	Database Design	Waterfall - Planning	Database Design Document
7	UI/UX Design	Waterfall - Planning	Wireframes and mockups for all key screens
8	Technology & Architecture Selection	Waterfall - Planning	Technology stack decision document
9	Development Planning → Product Backlog	Sign-off - Transition to Agile	Product Backlog (Epics → Features → User Stories → Acceptance Criteria)
10	Sprint Cycles (Agile / Scrum Development)	Agile / Scrum - Implementation	Working, tested software increment after every sprint
11	System-Wide Testing	Waterfall - Completion	Test Plan and Test Cases
12	Deployment	Waterfall - Completion	Live production system
13	Final Documentation	Waterfall - Completion	Final SRS and System Design Document
14	Maintenance & Continuous Improvement	Ongoing - post-launch	Bug-fix and update log